

FORT MCMURRAY, AB, Canada

WMO: 716890

Lat: 56.6500N

Lon: 111.2208W

Elev: 369

StdP: 96.97

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.7	(c) -32.9	(d) -40.5	(e) 0.1	(f) -35.6	(g) -37.5	(h) 0.1	(i) -32.7	(j) 9.3	(k) -5.0	(l) 8.0	(m) -6.9	(n) 1.0	(o) 100	(p) 0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.4	(c) 28.9	(d) 17.4	(e) 26.9	(f) 16.5	(g) 25.1	(h) 15.7	(i) 18.9	(j) 25.9	(k) 17.8	(l) 24.2	(m) 16.9	(n) 22.9	(o) 3.9	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.5	12.3	20.9	15.4	11.5	20.0	14.4	10.7	19.1	55.1	26.3	51.8	24.3	48.8	22.9	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.5	7.4	6.4	-39.5	32.5	2.4	1.6	-41.2	33.7	-42.5	34.7	-43.9	35.6	-45.6	36.8
			-39.6	20.9	2.3	1.4	-41.2	21.9	-42.6	22.8	-43.9	23.6	-45.6	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-16.8	-13.5	-6.6	2.7	9.9	15.0	17.5	15.6	10.2	2.9	-7.4	-14.7	(6)	
(7)		DBStd	13.66	9.40	8.15	8.52	5.88	4.87	3.50	2.88	3.51	4.48	4.88	6.86	8.70	(7)	
(8)		HDD10.0	3862	831	657	516	224	61	4	0	3	51	226	523	767	(8)	
(9)		HDD18.3	6263	1089	891	774	468	262	109	51	96	246	479	773	1025	(9)	
(10)		CDD10.0	688	0	0	0	6	58	154	231	177	57	5	0	0	(10)	
(11)		CDD18.3	49	0	0	0	0	2	9	24	12	1	0	0	0	(11)	
(12)		CDH23.3	825	0	0	0	3	60	178	326	213	44	2	0	0	(12)	
(13)		CDH26.7	189	0	0	0	0	13	37	78	51	10	0	0	0	(13)	
(14)	Wind	WSAvg	2.9	2.5	2.6	3.0	3.3	3.2	2.9	2.7	2.7	2.9	3.0	2.8	2.5	(14)	
(15)	Precipitation	PrecAvg	476	20	16	18	23	40	73	83	72	53	31	25	23	(15)	
(16)		PrecMax	724	40	50	46	52	109	188	147	177	132	88	57	42	(16)	
(17)		PrecMin	242	5	3	4	0	9	28	19	7	12	0	8	6	(17)	
(18)		PrecStd	98	9	10	10	13	24	35	34	40	31	20	13	9	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	7.3	12.8	22.8	28.6	30.0	31.1	30.8	28.2	21.1	9.8	5.8	(19)	
(20)			MCWB	2.9	2.8	5.5	11.3	14.9	16.8	19.0	18.7	16.5	12.2	4.8	2.4	(20)	
(21)		2%	DB	3.2	3.8	9.7	18.3	24.7	27.4	28.9	27.9	23.9	15.7	5.9	2.5	(21)	
(22)			MCWB	1.0	0.6	3.8	8.7	13.0	15.5	18.1	17.6	15.0	9.1	2.3	0.4	(22)	
(23)		5%	DB	0.2	0.8	7.2	15.3	22.3	25.3	26.9	25.7	21.1	12.9	3.4	-0.1	(23)	
(24)			MCWB	-1.3	-1.5	2.4	7.0	11.8	14.6	17.0	16.8	13.4	7.6	1.0	-1.6	(24)	
(25)		10%	DB	-3.4	-1.8	4.8	12.7	19.9	23.3	24.9	23.6	18.5	10.3	1.4	-2.8	(25)	
(26)			MCWB	-4.3	-3.2	1.2	5.6	10.6	13.9	16.4	15.8	12.3	6.1	-0.1	-3.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.1	5.9	11.4	16.0	18.7	20.6	20.4	17.4	12.7	5.3	2.7	(27)	
(28)			MCDWB	5.7	6.8	12.1	21.9	24.9	26.9	27.1	28.2	25.4	19.2	8.6	5.1	(28)	
(29)		2%	WB	1.1	0.9	4.1	9.0	14.2	17.0	19.1	18.7	15.7	9.7	2.9	0.5	(29)	
(30)			MCDWB	3.1	3.7	9.1	17.5	22.7	23.5	26.0	25.6	22.7	14.5	5.3	2.2	(30)	
(31)		5%	WB	-1.3	-1.3	2.7	7.4	12.7	15.9	18.2	17.6	14.1	8.0	1.3	-1.5	(31)	
(32)			MCDWB	0.2	0.8	6.8	14.6	20.4	22.0	24.5	23.9	19.8	12.2	3.1	-0.1	(32)	
(33)		10%	WB	-4.3	-3.3	1.3	6.0	11.2	15.0	17.2	16.5	12.8	6.4	0.0	-3.9	(33)	
(34)			MCDWB	-3.4	-1.8	4.6	12.0	18.5	21.3	23.2	22.3	17.3	10.0	1.3	-2.8	(34)	
(35)	Mean Daily Temperature Range			MDBR	9.5	11.7	12.6	12.3	13.6	12.7	12.4	12.5	12.0	9.1	7.9	8.8	(35)
(36)		5% DB	MCDBR	12.6	14.4	14.2	17.0	17.6	16.9	16.3	16.7	16.9	14.3	10.3	11.2	(36)	
(37)			MCWBR	10.8	11.6	9.4	9.6	8.5	7.7	7.5	8.2	9.2	8.7	7.9	9.7	(37)	
(38)		5% WB	MCDWB	12.3	13.9	13.4	16.1	15.6	13.1	13.8	14.8	15.1	13.3	9.3	11.1	(38)	
(39)			MCWBR	10.7	11.4	9.0	9.5	7.9	6.7	7.2	7.9	8.6	8.4	7.4	9.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.239	0.261	0.282	0.325	0.342	0.365	0.377	0.355	0.319	0.288	0.261	0.227	(40)	
(41)		taud		2.296	2.333	2.377	2.415	2.434	2.377	2.403	2.432	2.532	2.578	2.382	2.224	(41)	
(42)		Ebn at Noon		688	820	891	896	897	876	856	854	844	792	654	592	(42)	
(43)		Edh at Noon		47	70	87	99	104	111	107	97	76	56	44	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg		0.69	1.68	3.27	4.68	5.56	5.62	5.41	4.46	3.06	1.60	0.68	0.44	(44)	
(45)		RadStd		0.06	0.09	0.20	0.33	0.56	0.46	0.29	0.33	0.30	0.23	0.08	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page